

Information loss of fixed linear health indicators under competing degradation mechanisms and variable operating conditions

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Abstract

Health indicators combine condition-monitoring channels through weights fixed at design time, whereas the combination most informative about remaining useful life, the informative direction, rotates when degradation mechanisms progress at different rates. This paper quantifies the information a fixed linear indicator discards and whether the operating point, which governs those rates, can reduce it. For units differing by a time shift, the loss depends only on the angle between the indicator and the informative direction, and inflates the error variance of the remaining-life estimate by an exact factor. If this direction traverses an arc Φ over life, the best fixed indicator inflates the variance by at most $1/\cos^2(\Phi/2)$; on 129 measured battery cells the arc is 30.7° , so by at most 7.5%. On those cells a remaining-life estimator and a replacement policy perform as well with that indicator as with all channels. In simulation, steering the operating point reduces the information loss by a factor of 47, yet a practical estimator using the indicator stays within 1.2% of an estimator using every channel, while lifetime varies 19-fold. Unless the direction rotates widely, a fixed indicator is adequate, and the operating point is better selected for lifetime.

Keywords: health indicator, remaining useful life, prognostics, predictive maintenance, competing degradation mechanisms, operating conditions, information loss

1. Introduction

Condition monitoring seldom supplies raw measurements directly to a prognostic algorithm. Channels such as vibration, temperature, impedance and capacity are first combined into a scalar health indicator, and the prognostic algorithm operates on that scalar alone [1, 2]. The weights of the combination are usually fixed at the design stage, because they are embedded in firmware, frozen by certification, or impractical to re-tune in the field. The indicator therefore determines how accurately the remaining useful life can be estimated and, through that estimate, how conservatively maintenance must be scheduled.

Engineering components commonly degrade through several mechanisms at once, and these mechanisms need not progress at the same rate: one may dominate early in life and saturate, while another accelerates towards failure. Consequently, the combination of channels that carries the most information about the remaining useful life rotates as the component ages, and a fixed indicator can be aligned with it at a single instant only. Information discarded by the indicator cannot be recovered by any subsequent prognostic algorithm [3].

Two questions of practical relevance therefore arise. The first concerns the magnitude of the information lost by a

fixed indicator, and whether it can be assessed before the indicator is designed. The second concerns the operating conditions (temperature, load or duty cycle). These vary, either because the duty imposes them or because the operator selects them, and in both cases they alter the relative rates of the mechanisms; the question is whether a selected operating point can keep the indicator aligned and thereby reduce the loss.

Existing work addresses the first question from the perspective of indicator construction, and its criteria differ from the one used here. Health indicators are constructed from data against criteria such as monotonicity and trendability [2, 4], or by fusing sensors into a composite index whose weights are fitted once, for example so that the fused index has a consistent value at failure or follows a prescribed degradation model [5, 6, 7, 8, 9]; such indices have also been built under several operating conditions [10] and learned by neural networks [11], and reviews survey the resulting pipelines [1, 12]. Once fitted, these mappings are fixed and linear in the channels, which is precisely the class of indicators considered in this paper, but none of the construction criteria measures the information about remaining life that the index discards as the mechanisms rotate. State-space fusion [13] avoids a fixed mapping but tracks a single degradation state, so that no combination of channels can rotate. Section 4 compares one fusion criterion with the design proposed here on measured cells.

The second question connects to degradation modelling

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and to operation. Degradation under time-varying operating conditions is well modelled [14, 15, 16, 17, 18, 19, 20], and several degradation characteristics are known to respond differently to the same environment [21] or to depend on one another [22]. A separate body of work selects the operating point, or the load or production rate it implies, to extend life or to balance performance against reliability [23, 24, 25, 26, 27, 28], and prognostic accuracy has been assessed through its consequences for predictive maintenance [29]. These studies treat the monitoring signal as given; none asks what the operating point does to the information a fixed indicator retains. Our previous work bounded the information that any fixed compression retains along a given trajectory [3] and the number of run-to-failure units required to learn such a compression [30]; both analyses treat the trajectory as fixed.

The main contributions of this paper are as follows.

- The information discarded by a fixed linear indicator is shown to be an exact function of a single misalignment angle, and the resulting inflation of the error variance of the remaining-life estimate, and of the safety margin of a predictive replacement policy, is obtained in closed form (Section 2).
- The arc traversed by the informative direction over the service life is shown to bound this inflation before any indicator is designed, which yields a five-step design procedure that requires run-to-failure data only (Section 3).
- In a case study on measured battery cells the bound is 7.5%, which corresponds to a safety margin of a predictive replacement policy at most 3.7% wider; a leave-one-out remaining-life estimator and a replacement policy using the fixed indicator perform as well as ones using all channels (Section 4).
- Selecting the operating point to keep the indicator aligned reduces the information loss substantially but leaves the error of a practical remaining-life estimator essentially unchanged, whereas the operating point alters the lifetime by an order of magnitude (Section 5). Public run-to-failure data support the rotation premise but do not permit a test of the effect of the operating point on degradation (Section 6).

The remainder of the paper is organised as follows. Section 2 formulates the degradation model and the information loss. Section 3 derives a bound on the loss in terms of the rotation arc and the design procedure that follows from it. Section 4 applies the procedure to measured lithium-ion cells. Section 5 examines whether selecting the operating point is worthwhile, and Section 6 examines the premises of the model on public run-to-failure data. Section 7 discusses the implications and concludes the paper. Derivations are given in Appendix A.

2. Problem formulation

2.1. Degradation model with operating-point dependence

A component degrades through d mechanisms, with damage states $x = (x_1, \dots, x_d)$, $x_i \geq 0$. A single operating point $u(t) \in \mathcal{U} \subset \mathbb{R}^m$, comprising variables such as temperature, load and duty cycle within a compact envelope $\mathcal{U}$, drives all of them:

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d. \quad (1)$$

The stress factor $\varphi_i(u) > 0$ expresses the acceleration of mechanism i at u relative to a reference operating point, and the shape function $h_i(x_i) > 0$ describes the dependence of its rate on the damage already accumulated. This is the standard accelerated-life form [14]: in the Arrhenius model $\varphi_i(u) = A_i \exp(-E_i/(kT))$ with activation energy E_i , and suitable choices of h_i yield exponential, power-law and stretched-exponential degradation [3]. Damage is normalised such that each mechanism exhausts the component at unity, and the component is withdrawn from service at a condemnation limit $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (2)$$

2.2. Remaining useful life and the informative direction

The quantity of interest is the remaining useful life $R(t) = T_f - t$. For a single unit with known parameters, (1) would determine it exactly; the uncertainty arises from variability between units [14]. The units of the population are assumed to differ in their position along a common degradation path, that is, by a time shift, and R is taken to be Gaussian over the population with variance $\text{Var}[R]$. Since R decreases at unit rate, the sensitivity of mechanism i to the remaining life is

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (3)$$

Populations whose units differ in their degradation coefficients rather than in their timing are not covered: for the system of Section 5.2, dispersing the coefficients rotates the regression of damage on remaining life by about 45° relative to (3) (Supplementary Section S4). Section 4 examines the practical consequence on measured cells whose lives differ more than fourfold.

Each mechanism is observed through a dedicated channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (4)$$

with independent standard normal noise ϵ_i . This pairing of channels with mechanisms is an assumption: condition monitoring in general yields $y = H(x, u) + \epsilon$, with sensors that mix mechanisms, and Section 6 shows that raw turbofan sensors violate it. Whitening each channel by its noise level gives the signal vector and its direction,

$$\mathbf{d} = [\sigma_i^{-1} \varphi_i(u) h_i(x_i)]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|. \quad (5)$$

The norm $\|\mathbf{d}\|$ measures the sensitivity of the record to the remaining life, and $\hat{\mathbf{d}}$ is the combination of channels along which this sensitivity acts, referred to as the *informative direction*. All entries of $\mathbf{d}$ are positive, and $\hat{\mathbf{d}}$ depends on both the state and the operating point. The operating point affects $\hat{\mathbf{d}}$ through two paths: indirectly, through the damage it produces, and directly, through $\varphi_i(u)$ (Fig. 1).

2.3. Information loss of a fixed linear indicator

A fixed linear indicator maps the whitened record $\tilde{y} = \Sigma^{-1/2}y$, $\Sigma = \text{diag}(\sigma_i^2)$, through a weight vector selected once at the design stage:

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1. \quad (6)$$

An indicator defined on raw measurements is the same mapping with $\mathbf{v} \propto \Sigma^{1/2}\mathbf{w}$. Let ψ denote the angle between $\mathbf{v}$ and $\hat{\mathbf{d}}$, and $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ the signal-to-noise ratio of the record. Linearising the record about the nominal path renders it jointly Gaussian with R ; the indicator retains the fraction $\cos^2 \psi$ of the signal and all of the noise. The information about R that it discards relative to the complete record is therefore (Appendix A.1)

$$\ell = \frac{1}{2} \log \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (7)$$

with $w = \gamma/(1 + \gamma)$, and the error variance of the optimal estimate of R based on the indicator exceeds that based on the complete record by the factor

$$\frac{\text{Var}[R | z]}{\text{Var}[R | \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}. \quad (8)$$

The loss thus depends on the indicator only through a single angle, and for small misalignment $\ell \approx \frac{w}{2}\psi^2$. The design objective is its average over the service life,

$$\bar{\ell} = \frac{1}{T_f} \int_0^{T_f} \ell(t) dt. \quad (9)$$

The same reduction applies to the information discarded over a complete record, with γ summed over the measurement instants and $\sin^2 \psi$ averaged with weights $\|\mathbf{d}\|^2$ (Supplementary Section S2).

The factor (8) has a direct consequence for maintenance. Under the Gaussian posterior of the model, a predictive replacement policy that replaces a unit once its probability of failing within a lead time h reaches α does so when the estimated remaining life falls to $h + z_{1-\alpha}\sigma$, where σ is the posterior standard deviation and $z_{1-\alpha}$ the standard normal quantile. The safety margin $z_{1-\alpha}\sigma$ is, on average, remaining life discarded at replacement beyond the lead time, and a fixed indicator multiplies it by e^ℓ relative to the complete record. Decision-oriented assessments of prognostic algorithms weigh precisely this margin against the risk of failure [29].

Equation (7) also identifies the relevant Fisher information. The design of accelerated degradation tests and of

inputs for estimability usually maximises the Fisher information of the record [31], here $I_{\text{rec}} = \|\mathbf{d}\|^2$ per unit prior variance about R ; the loss, however, is determined by this quantity together with the Fisher information of the indicator itself, $I_z = (\mathbf{v}^\top \mathbf{d})^2$:

$$\ell = \frac{1}{2} \log(1 + \text{Var}[R] I_{\text{rec}}) - \frac{1}{2} \log(1 + \text{Var}[R] I_z). \quad (10)$$

For a fixed value of I_{rec} the loss still ranges from zero to $\frac{1}{2} \log(1 + \gamma)$ as ψ varies. Maximising the information in the record therefore does not minimise the loss of a fixed indicator, whereas including I_z in the criterion does. Supplementary Section S4 illustrates this on the system of Section 5.2.

3. Rotation of the informative direction and a bound on the loss

3.1. Rotation rate of the informative direction

Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i and $p_i = \hat{\mathbf{d}}_i^2$ the share of the record's Fisher information carried by channel i . Differentiating (5) along (1) gives (Appendix A.2)

$$\|\dot{\mathbf{d}}\| = \text{Std}_p[\dot{g}], \quad \dot{g}_i = \varphi_i(u) h'_i(x_i) + \nabla_u \log \varphi_i(u) \cdot \dot{u}. \quad (11)$$

The informative direction thus rotates at a speed equal to the information-weighted spread of the logarithmic rates of the mechanisms. If all mechanisms accelerate in proportion, the direction remains fixed however rapidly the component degrades; rotation arises only from differences between the mechanisms. The second term of $\dot{g}_i$ represents the influence of the operating point: a change in the operating point alters the rates that rotate the direction. Only the component of this rotation along the great circle towards $\mathbf{v}$ changes ψ ; the remaining $d-2$ components rotate the direction without affecting the information retained by the indicator (Fig. 2(b)).

3.2. A bound in terms of the rotation arc

Suppose that the informative direction traverses a total arc $\Phi < \pi$ over the service life. The direction at the midpoint of this arc lies within $\Phi/2$ of every point on it, so a fixed indicator placed at the midpoint satisfies $\psi \leq \Phi/2$ throughout. Since the loss increases with ψ , the best fixed indicator satisfies, at every instant,

$$e^{2\ell} \leq \frac{1}{1 - w \sin^2(\Phi/2)} \leq \frac{1}{\cos^2(\Phi/2)}. \quad (12)$$

The bound requires only the arc, which can be estimated from run-to-failure data before any indicator is designed. An arc of 30° allows at most 7.2% additional error variance, 60° at most 33%, and 90° a doubling; the right-hand bound corresponds to the limit of high signal-to-noise ratio and is independent of γ . Correspondingly, the safety margin of the replacement policy of Section 2.3 grows by

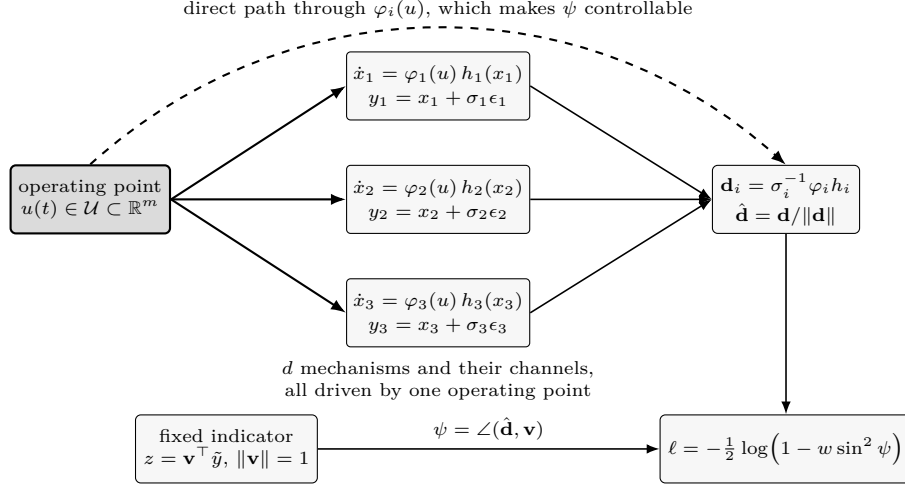


Figure 1: Overview of the problem formulation. The operating point affects the informative direction both through the damage and directly through the stress factors $\varphi_i(u)$; the direct path makes the misalignment ψ controllable. A single operating point drives all d mechanisms.

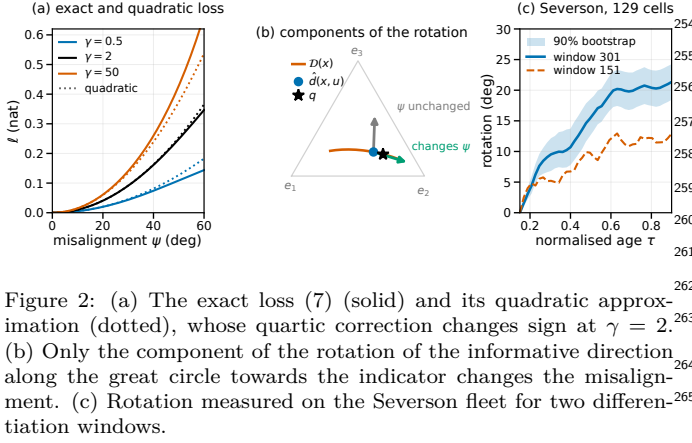


Figure 2: (a) The exact loss (7) (solid) and its quadratic approximation (dotted), whose quartic correction changes sign at $\gamma = 2$. (b) Only the component of the rotation of the informative direction along the great circle towards the indicator changes the misalignment. (c) Rotation measured on the Severson fleet for two different titration windows.

at most the factor $1/\cos(\Phi/2)$. The best fixed indicator lies close to the midpoint of the arc; Supplementary Section S2 characterises it as a barycentre of the path of the informative direction. For the system of Section 5.2 under constant inputs, the arc is between 24.8° and 35.6° , the bound allows between 4.8% and 10.3%, and the best fixed indicator for each input incurs between 1.6% and 3.4%; the bound therefore holds and overestimates the excess by a factor of about three.

Figure 2(a) shows the dependence of the loss on the misalignment angle, and Fig. 2(c) the rotation measured on a battery fleet (Section 4).

3.3. Design procedure

The results above yield a design procedure that requires run-to-failure data only.

1. Estimate the informative direction along normalised life as the time derivative of the monitoring channels, each whitened by its measurement noise, averaged over the fleet.

2. Compute the arc Φ that the direction traverses and the bound (12); if the bound is acceptable, a fixed indicator suffices.
3. Place the fixed indicator at the barycentre of the path, in practice the renormalised mean direction.
4. Validate the indicator with a leave-one-out remaining-life estimator, against the same estimator run on all channels.
5. Evaluate the maintenance decision that the estimate supports.

Section 4 carries out the five steps on measured lithium-ion cells.

4. Case study: lithium-ion cells

The procedure is applied to the lithium-ion cells of Severson et al. [32], cycled to failure under different fast-charging protocols. The monitoring channels are discharge capacity, internal resistance and mean cell temperature. The 129 cells that meet the selection criteria of the protocol in Supplementary Section S1 have lives from 447 to 1932 cycles; because the lives differ more than fourfold across protocols, the population is far from a time shift, and the case study also probes that assumption. Supplementary Section S1 gives the protocol and parameters of every step.

4.1. Rotation and the bound

The fleet-mean informative direction moves through an arc of 30.7° between 15% and 90% of life (Fig. 2(c)). By (12), the fixed indicator at its barycentre therefore inflates the error variance of the remaining-life estimate by at most 7.5% over that window, irrespective of the signal-to-noise ratio, and widens the safety margin of a predictive replacement policy by at most 3.7%. In whitened units the indicator weights capacity, resistance and temperature as

(0.99, 0.10, 0.12) and lies at most 14° from the informative direction.

4.2. Remaining-life estimation

A similarity-based estimator matches the last 30% of a cell's elapsed cycles against the fleet reference path in normalised life and returns the life that fits best; every-
thing it uses, including the fixed indicator, is computed without the cell being tested. With all three channels its root-mean-square error is 148, 83 and 49 cycles at 50%, 70% and 90% of life. Run on the fixed indicator at the barycentre of the measured direction, the same estimator attains between 0.96 and 1.00 times the mean squared error of the three-channel estimator at these ages (Table 1). Capacity alone, the conventional battery health indicator, lies up to 22° from the informative direction, for which the bound allows 17%, and raises the mean squared error by 5% at half of life and by less later. Temperature alone, a deliberately misaligned control, raises it at least 40-fold, so the test is able to detect a poorly chosen indicator. A data-level fusion indicator, whose weights minimise the between-cell spread of the indicator at failure relative to its mean change over life, in the spirit of [5], lies 75° to 82° from the informative direction and has 28.9 times the mean squared error at half of life. Near failure, the stage its construction targets, it is the most accurate signal, at 0.81 times the three-channel error: it suppresses between-cell differences in the level of the channels at failure, a source of error that the time-shift population of Section 2 does not contain.

4.3. Replacement decisions

The last step evaluates the decision directly. Each cell is inspected every 25 cycles from cycle 100 and is replaced at the first inspection at which the predicted remaining life is at most a threshold θ ; a cell not replaced before its end of life counts as a failure. For a failure ten times as costly as a preventive replacement, the threshold that minimises the long-run cost per cycle [29] gives the costs of Table 1. The fixed indicator, capacity alone and all three channels reach the same cost to within 0.1%, with 1.6% of cells failing and about 134 cycles, a fifth of a cell's life, left unused at replacement; the fusion indicator costs 2.5% more. For cost ratios from 5 to 50 the fixed indicator stays within 1% of the three-channel cost. The thresholds are chosen on the same cells for every signal, so the absolute costs are optimistic but the comparison is not. The case study thus confirms, on measured data and at the level of a maintenance decision, what the bound predicts: for this fleet a fixed indicator is adequate.

5. Selecting the operating point

The operating point sets the relative rates of the mechanisms and therefore moves the informative direction. This section asks whether selecting it so as to keep a fixed indicator aligned, referred to as steering, is worthwhile.

5.1. Limits of steering

At a state x , the directions attainable through the operating point form the set

$$\mathcal{D}(x) = \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\}, \quad (13)$$

and the smallest misalignment achievable at x is the floor $\beta_{\min}(x) = \angle(\mathbf{v}, \mathcal{D}(x))$. Steering has two structural limits, derived in Supplementary Section S4. Holding the misalignment at a constant, non-zero level is a single scalar condition and needs only one adjustable operating variable. Removing the loss altogether confines the state to a straight segment in damage space and, except in special cases, needs one fewer independent operating variable than there are mechanisms, together with an aligning input that remains admissible along that segment. A component with three mechanisms and a single temperature setpoint can therefore reduce the loss but, generically, not remove it.

When the loss cannot be removed, the relevant quantity is the smallest achievable average loss,

$$\bar{\ell}^* = \min_{u(\cdot)} \frac{1}{T_f} \int_0^{T_f} \ell \, dt. \quad (14)$$

Both the integral and the lifetime T_f depend on the operating point, so the objective is a ratio: the operating point that aligns the indicator is in general not the one that prolongs life. The simplest policy disregards the lifetime and applies, at each state, the input that minimises the misalignment; it is referred to as the myopic policy.

5.2. A numerical study

The study considers $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is a scaled reciprocal temperature, $u = 3(1/(kT) - 1/(kT_{60})) / (1/(kT_{25}) - 1/(kT_{60}))$ with T_{60} and T_{25} the absolute temperatures of 60 and 25°C , so that $u = 0$ corresponds to 60°C and $u = 3$ to 25°C . The parameters are $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, which in these units are activation energies of 0.44, 0.81 and 0.59 eV, with shape functions $h_i(x) = 1 + c_i x$ and $c = (-0.5, 2.0, 0.8)$, which give one decelerating and two accelerating mechanisms. Time is measured in units of the characteristic time of the first mechanism at 60°C , where $\varphi_1 = 1$. There is a single input $u \in [0, 3]$, so by Section 5.1 the loss can be reduced but, generically, not removed; Supplementary Section S4 confirms this for the present system. The noise has unit variance and the prior variance is $\text{Var}[R] = 10^3$, so the weight $w = \gamma/(1 + \gamma)$ of (7) is evaluated at the system's own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than held fixed. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$; the values of the reported constants at $x_f = 1$ are given in Supplementary Section S4. The identities used below were verified numerically (Supplementary Section S3).

The myopic policy applies, at each state, the input that minimises the misalignment, searched on a grid of

Table 1: Case study on the Severson cells. $\psi_{\max}$ is the largest angle between the signal and the fleet-mean informative direction over $\tau \in [0.15, 0.90]$; the MSE ratios compare the leave-one-out remaining-life estimator run on the signal with the same estimator run on all three channels, at 50%, 70% and 90% of life; the replacement columns give the minimal long-run cost per 1000 cycles, in units of a preventive replacement, when a failure costs ten times as much, with the fraction of cells failing and the mean life left unused at replacement, in cycles.

signal	$\psi_{\max}$	MSE ratio			replacement policy		
		50%	70%	90%	cost	failures	unused
all channels	—	1	1	1	1.689	1.6%	134
fixed indicator	14°	0.960	0.972	0.999	1.688	1.6%	133
capacity	22°	1.050	1.014	0.986	1.690	1.6%	134
fusion	82°	28.9	8.43	0.807	1.731	1.6%	150
temperature	90°	43.7	171	725	—	—	—

41 points of $\mathcal{U}$. Minimising its loss over the indicator yields the steering indicator q^* , with weight vector $\mathbf{v}^* = (0.466, 0.771, 0.435)$, at which the myopic policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input, a factor of 68. This comparison holds the indicator fixed at the one designed for steering. For operation without steering, the indicator would instead be designed for the constant input, and minimising over the indicator and the constant input jointly gives 7.95×10^{-3} , at the high-temperature end $u = 0$ of the envelope. Relative to this baseline, the reduction achieved by steering is a factor of 47. With control grids of 81 points the two factors become 72 and 50, and on 161 points 72 and 50 again. The myopic policy is also close to the best achievable by any policy: the fractional optimum of (14), computed on state grids of up to 137 points per axis, lies within 2.1% of it on every control grid (Supplementary Section S4).

Table 2 reports the practical consequences. By (8), a loss of 7.95×10^{-3} already corresponds to an error variance only 1.6% above that of the complete record, and steering removes almost all of this excess. A practical estimator, however, is largely insensitive to it. A time-shifted population is simulated along the path of each policy and observed at 30 instants over the last 40% of a window with unit channel noise, and the remaining life is estimated from the offset that best matches the reference path of the policy, once from the indicator and once from all three channels. Under every policy the two estimators differ by less than 1.2%, and the error as a fraction of life stays between 8.4% and 10.7%, being determined by the path rather than by the indicator. The operating point does, however, change the lifetime substantially. Across the constant designs of the table it varies 19-fold; the myopic loop at q^* attains a lifetime 1.66 times that of the best constant input at the same indicator and 3.8 times that of the constant design with the smallest loss; and a myopic loop aimed at a balanced indicator, which requires far less steering, extends the lifetime by a further factor of 3.4.

5.3. Sensitivity to model misspecification

An indicator designed from a model inherits the errors of that model. For each level ε , a perturbed energy vector $E(1 + \varepsilon\eta)$ is drawn, with η standard normal, and the

barycentre indicator is constructed from this misspecified system along the trajectory of a constant input, as the design procedure of this paper prescribes. The indicator is then evaluated on the true system under the myopic policy, with every loss evaluated at the system's own γ . Since the policy is computed from the true system, the table isolates the cost of an error in the indicator; an error that also affected the controller would be more costly. Table 3 reports the ratio of this loss to that of the indicator designed from the true model, over 40 draws per level; a ratio of 1 indicates that the error has no cost.

An error of 2% in the energies has a negligible cost. At 5% the median cost is a factor 1.40. At 10% the median cost is 3.10 and the worst of forty draws costs 67.7, comparable in magnitude to the entire benefit of steering; beyond this level every column increases monotonically, to a median of 36.0 at $\varepsilon = 0.35$. The loss to which these ratios apply is small, however, so that even a tenfold increase leaves the indicator's excess error variance below 1%.

6. Evidence from public run-to-failure data

Three premises of the model can be tested on public run-to-failure data: that the informative direction rotates; that the operating point moves it by acting on the degradation; and that the separable stress model, interpreted through the pairing of channels with mechanisms, holds. Table 4 summarises what six public run-to-failure fleets establish and what they leave open. Supplementary Section S1 gives the protocol behind every row, and its Table S1 lists the parameters of the code that produced it.

Rotation of the informative direction is observed on the battery cells of Section 4. That analysis and the turbofan contrasts below interpret a time derivative as the sensitivity (3), and therefore presuppose the time-shift population of Section 2.

The effect of the operating point on degradation remains untested. The measured fleets either do not vary the operating point within a unit or contain too few units. The two turbofan simulations do vary it, but their generators preclude the test. C-MAPSS imposes its flow and efficiency losses as exponential functions of time, from a random initial deterioration, and lets the regime act only on the thermodynamic response of the sensors [36]. N-CMAPSS imposes a linear normal degradation followed

Table 2: Information loss, remaining-life estimation error and lifetime of the test system under different operating strategies. $\bar{\ell}$ is the information loss (9); $e^{2\bar{\ell}} - 1$ the corresponding excess error variance (8); the RUL errors are the median absolute error of a template-matching remaining-life estimator over 2000 units, as a percentage of life, based on the indicator or on all three channels; T_f is the lifetime. Constant designs use the indicator that is best for their own input.

operating strategy	$\bar{\ell}$	$e^{2\bar{\ell}} - 1$	RUL error (% of life)		
			indicator	all channels	T_f
constant, 60 °C	7.95×10^{-3}	1.6%	10.7%	10.8%	0.378
constant, 41.5 °C	1.44×10^{-2}	2.9%	8.9%	8.8%	1.967
constant, 25 °C	1.26×10^{-2}	2.6%	8.8%	8.7%	7.203
myopic, q^*	1.68×10^{-4}	0.03%	9.5%	9.5%	1.433
myopic, balanced	4.08×10^{-3}	0.8%	8.4%	8.5%	4.928

Table 3: Cost of designing the indicator from a misspecified stress model. Entries are the loss on the true system relative to that of the indicator designed from the true model, over 40 draws per level; ε is the relative standard deviation of the error in the activation energies. The steering policy uses the true system.

ε	median	90th percentile	worst of 40
0.02	1.03	1.52	2.04
0.05	1.40	2.94	6.44
0.10	3.10	10.8	67.7
0.20	10.0	44.9	78.6
0.35	36.0	94.1	195

7. Discussion and conclusions

The results have two practical implications.

First, the rotation arc should be assessed before an adaptive indicator is considered. The loss of a fixed indicator is governed by the extent to which the informative direction rotates, and (12) converts the arc into a worst-case increase in the error variance of the remaining-life estimate. For the battery fleet examined here the resulting increase is at most 7.5%, and a remaining-life estimator and a replacement policy using the fixed indicator perform as well there as ones using all channels; for the synthetic system the actual excess is about one third of the bound. Where the arc amounts to a few tens of degrees, a single fixed indicator is adequate; adaptive, segmented or learned indicators become worthwhile as the arc approaches a right angle, at which the bound allows the error variance to double. The five steps of Section 3.3 make this assessment from run-to-failure data alone.

Second, the operating point should be selected primarily for service life. Steering the operating point can reduce the information loss by an order of magnitude or more. Where the loss is already small, however, this yields almost no improvement in the accuracy of a practical remaining-life estimator, whereas the same choice alters the lifetime by an order of magnitude. Aligning an indicator and prolonging the life of the component are distinct objectives, and the latter is usually the more consequential.

Limitations. Four limitations qualify these conclusions. First, the construction assumes a time-shift population, whereas dispersing the degradation coefficients of the test system rotates the population regression of damage on remaining life by about 45° from the direction the construction uses, more than the measured arc of the battery cells, so that the bound does not cover such dispersion even though the battery case study suggests that the practical conclusion survives it. Second, the pairing of channels with mechanisms fails on raw turbofan sensors, so the geometry applies to them only after the observation map has been inverted. Third, the actuation premise remains untested on every public fleet. Finally, the best achievable loss is computed and bracketed between the myopic policy and an extrapolation, not bounded analytically.

by a steeper abnormal one, with the operating history entering through the onset of the abnormal phase [38], and the health parameters supplied with the data are constant within every flight. The C-MAPSS contrast, positive in all but three engines against a negative control that returns none, therefore shows that the observed direction depends on the regime through the sensor response alone. Variable operating conditions thus rotate the observed direction even when they do not act on the damage. Because the loss of a fixed indicator depends on the observed direction regardless of the cause of its motion, this finding is relevant to indicator design; it is not, however, evidence that the operating point acts on the damage.

The separable model fails on raw turbofan sensors. On both N-CMAPSS datasets the interaction between operating point and life stage exceeds the main effect of the operating point, and since the simulated damage does not respond to the operating point within a flight, the interaction is attributable to the sensor map: the sensors mix mechanisms with weights that depend on the operating point, in violation of (4). The pairing is what makes the informative direction positive, so the geometry of Sections 3 and 5 applies to such sensors only after the observation map has been inverted, and the offline computation of the actuation authority described in Supplementary Section S4 is available only where the separable form has been shown to hold. The two-way layout, whose protocol is given in Supplementary Section S1, provides this test at the cost of a single pass over data that a reliability programme typically already holds.

Table 4: Premises of the model examined on six public run-to-failure fleets. Only the two turbofan simulations vary the operating point within a unit, and neither allows it to act on the damage; the actuation premise is therefore untested on every public fleet.

fleet	premise	quantity measured	finding
Severson lithium-ion [32]	rotation	arc of an empirically constructed informative direction, the fleet mean, over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 2(c)); remains in the open positive orthant once the window reaches 301 cycles
XJTU-SY [33], PRONOSTIA [34]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated
NASA batteries [35]	actuation	—	only four cells
C-MAPSS [36]	actuation: not testable	between-regime minus within-regime angle, each engine serving as its own control	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; surrogate regimes give $ B - W \leq 0.5^\circ$; the damage is independent of the regime by construction, so the contrast reflects the sensor map
N-CMAPSS DS02, DS01 [37]	separability with pairing (4)	interaction against both main effects, in a two-way layout inside engines	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected jointly

Appendix A. Derivations

Appendix A.1. Information loss and error-variance inflation

Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$, the linearised whitened record. The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$. The record’s sufficient statistic $\mathbf{d}^\top \tilde{y}$ has ratio γ . A scalar Gaussian channel of signal-to-noise ratio s carries $\frac{1}{2} \log(1+s)$ nats about R [39], and subtracting gives the first form of (7); substituting $\cos^2 \psi = 1 - \sin^2 \psi$ gives the second. The posterior variances are $\text{Var}[R]/(1 + \gamma \cos^2 \psi)$ and $\text{Var}[R]/(1 + \gamma)$, whose quotient is (8). Equation (10) is (7) with $\gamma = \text{Var}[R]I_{\text{rec}}$ and $\gamma \cos^2 \psi = \text{Var}[R]I_z$; at fixed I_{rec} , w is fixed and ℓ increases continuously from zero towards $\frac{1}{2} \log(1 + \gamma)$ as ψ goes from 0 to $\pi/2$.

Appendix A.2. Rotation of the informative direction

From $\mathbf{d}_i = e^{g_i}$, $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, and $\dot{\mathbf{d}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \dot{\mathbf{d}}(\dot{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has components $\dot{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$ with $p_i = \dot{\mathbf{d}}_i^2$, so its squared norm is $\text{Var}_p[\dot{g}]$. Differentiating $g_i = \log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (1) gives $\dot{g}_i$ in (11). With $\cos \psi = \dot{\mathbf{d}}^\top \mathbf{v}$, $-\sin \psi \dot{\psi} = \dot{\mathbf{d}}^\top \mathbf{v}$, so only the component of $\dot{\mathbf{d}}$ along the great circle towards $\mathbf{v}$ changes ψ .

Data availability and reproducibility

The six fleets named in Section 6 are publicly available. One script recomputes every reported number from a single configuration and compares it with the value stated in this manuscript and in its supplementary material, covering the tables and the quantities reported alongside them; all of its checks pass, and the output of the run that verified this version is included. The random-system sweep and the downstream repetition of Supplementary Section S4 and the grid sequence of Section 5.2 require several hours of computation, so they are recorded by

three further scripts; the checks read these records and recompute part of the grid sequence afresh. One group of numbers lies outside the checks: the residuals of the identity checks in Supplementary Section S3. The figures are produced by three further scripts. All of these are available at <https://github.com/HuyHoangLe0201/health-indicator-information-loss>, and are also provided with this article as a supplementary archive, together with the Supplementary Material.

CRedit authorship contribution statement

Huy Hoang Le: Conceptualization, Methodology, Formal analysis, Software, Validation, Investigation, Data curation, Visualization, Writing – original draft. **Kim-Anh Nguyen:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with drafting and editing the text and with writing and checking the analysis and verification code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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